Innhold

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3.1 Definitions and examples

Definisjon 1: 3.1.1

A metric space (X, d) consists of a nonempty set X and a function

$$d : X \times X \to [0, \infty)$$

such that

- $\forall x, y \in X$ we have $d(x, y) \geqslant 0$ and $d(x, y) = 0 \iff x = y$
- $\forall x, y \in X$ we have $d(x, y) = d(y, x)$
- $\forall x, y, z \in X$ we have $d(x, y) \leqslant d(x, z) + d(z, y)$

A function d statisfying these conditions is called a metric on X

Oppgave 1: 3.1.1

We shall now take a look at an example of a different kind. Assume that we want to send messages in a language with N symbols (letters, numbers, punctuation marks, space, etc.). We assume that all messages have the same length K (if they are too short or too long, we either fill them out or break them into pieces). We let X be the set of all messages, i.e., all sequences of symbols from the language of length K. If x = (x1,x2,...,xK) and y = (y1,y2,...,yK) are two messages, we define

$$d(x, y) = \text{the number of indices n such that } x_n = y_n.$$

It is not hard to check that d is a metric. It is usually referred to as the Hamming- metric, and is much used in communication theory, where it serves as a measure of how much a message gets distorted during transmission.

Bevis 1: 3.1.1

This set X can be called a metric space if it satisfies the three conditions of def 3.1.1. We have that the distance between x and y is the number of indices n such that $x_n \neq y_n$. We have that d satisfies the first condition if and only if $x_n = y_n$ for all indeces n. This means that $x = y$ where

$$x = (x_1, x_2, ..., x_K)\, y = (y_1, y_2, ..., y_K)$$

And this is equivalent with $d(x, y) = 0$. Since $x = y \Rightarrow d(x, y) = d(x, x) = 0$

We now prove the second condition. $\forall x, y \in X$, $d(x, y) = d(y, x)$.

If $d = \#$ of indices s.t. $x_n \neq y_n$, we have that

$$d(x, y) = \# \text{ of indices such that } x_n \neq y_n$$

which is equivalent with

$$d(y, x) = \# \text{ of indices such that } y_n \neq x_n$$

So $d(x, y) = d(y, x)$

Now for the third condition. $\forall x, y, z \in X$ we have $d(x, y) \leqslant d(x, z) + d(z, y)$.

$$\begin{aligned} x &= (x_1, x_2, ..., x_K) \\ y &= (y_1, y_2, ..., y_K) \\ z &= (z_1, z_2, ..., z_n) \end{aligned}$$

Then

$$d(x, y) = \# \text{ of indices s.t. } x_n \neq y_n$$

$$d(x, z) = \# \text{ of indices s.t. } x_n \neq z_n d(z, y) = \# \text{ of indices s.t. } z_n \neq y_n$$

We know that if $d(x, z) = k, d(z, y) = r$ and $g = d(x, y)$, then $r + k \geqslant g$ and $r + k = g \quad \Longleftrightarrow \quad x = y$.

Oppgave 2: 1.3.2

Prove that $(\mathbb{R}^n, d)$ is a metric space, with

$$d(x, y) = \|x - y\| = \sqrt{\sum_{n}^{i=1} (x_i - y_i)^2}$$

Bevis 2: 1.3.2

- The first condition is that $d(x,y) \geqslant 0$ and that $d(x,y) = 0 \iff x = y$. We know that $d(x,y)$ is obviously larger than 0, and

$$d(x,x) = \sqrt{\sum_{n}^{i=1} (x_i - x_i)^2} = \sqrt{\sum_{n}^{i=1} 0} = 0$$

- The second condition is that $d(x,y) = d(y,x)$.

$$\begin{aligned} d(x,y) &= \sqrt{\sum_{n}^{i=1} (x_i - y_i)^2} \\ &= \sqrt{\sum_{n}^{i=1} (-1)^2 (y_i - x_i)^2} \\ &= d(x,y) \end{aligned}$$

- For the third condition, we have that

$$\begin{aligned} d(x,y) &= \|x - y\| \\ &= \|(x-z) + (z-y)\| \leqslant \|x-z\| + \|z-y\| \\ &= d(x,z) + d(z,y) \end{aligned}$$

Proposisjon 1: 3.1.4

$\forall x, y, z \in X$ with (X,d) metric space,

$$|d(x,y) - d(x,z)| \leqslant d(y,z)$$

Bevis 3: 3.1.4

We have that $d(x,y) \leqslant d(x,z) + d(z,y)$. Then $d(x,y) - d(x,z) \leqslant d(y,z)$. Now we need to prove the negative.

$$d(x,z) \leqslant d(x,y) + d(y,z) \Rightarrow d(x,z) - d(x,y) \leqslant d(y,z)$$

and thus

$$d(x,y) - d(x,z) = d(x,z) - d(x,y) \leqslant d(y,z) \Rightarrow |d(x,y) - d(x,z)| \leqslant d(y,z)$$

Oppgave 3: 3.1.6

If V is a vector space over R or C, a function $\|\cdot\| : V \to \mathbb{R}$ is called a norm if the following conditions are satisfied:

- (i) $\forall x \in V, \|x\| \geqslant 0$ and $\|x\| = 0 \iff x = 0$
- $\|\alpha x\| = |\alpha| \, \|x\| \quad \forall x \in V$ and $\alpha \in \mathbb{K}$
- $\|x + y\| \leqslant \|x\| + \|y\| \quad \forall x, y \in V$

Bevis 4: 3.1.6

- Since $\|\cdot\| : V \to \mathbb{R}$, then it satisfies all three conditions for a norm. To show that it satisfies the first condition of a metric, then

$$\|x\| = d(x, 0) = \|x - 0\| \geqslant 0$$

  and this becomes similar to the prove for norm

- Have to now show that $d(x, y) = d(y, x)$

$$\begin{aligned} d(x, y) &= \|x - y\| \\ &= \|(-1)(y - x)\| \\ &= |(-1)| \, \|y - x\| \\ &= \|y - x\| = d(y, x) \end{aligned}$$

- Have to now show that $d(x, y) \leqslant d(x, z) + d(z, y)$

$$\begin{aligned} d(x, y) &= \|x - y\| \\ \|(x - z) + (z - y)\| \leqslant \|x - z\| + \|z - y\| \\ &= d(x, z) + d(z, y) \end{aligned}$$

  And the proof is finished

Oppgave 4: 3.1.7

Show that if $(x_1, x_2, ..., x_n)$ are points in a metric space, then

$$d(x_1, x_n) \leqslant d(x_1, x_2) + d(x_2, x_3) + ... + d(x_{n-1}, x_n)$$

Bevis 5: 3.1.7

We use induction to prove this. Assume that

$$d(x_k, x_{k+2}) \leqslant d(x_k, x_{k+1}) + d(x_{k+1}, x_{k+1})$$

Then for $k = 0$, we get that

$$d(x_1, x_3) \leqslant d(x_1, x_2) + d(x_2, x_3)$$

For $k = k$ we assume that

$$d(x_k, x_{k+2}) \leqslant d(x_k, x_{k+1}) + d(x_{k+1}, x_{k+1})$$

and now we apply our hypothesis, that

$$d(x_k, x_{k+3}) \leqslant d(x_k, x_{k+2}) + d(x_{k+2}, x_{k+3})$$

since $d(x_k, x_{k+2}) \leqslant d(x_k, x_{k+1}) + d(x_{k+1}, x_{k+1})$, we do get that

$$d(x_k, x_{k+3}) \leqslant d(x_k, x_{k+2}) + d(x_{k+2}, x_{k+3})$$

and this will now apply for all k. Thus the statement that

$$d(x_1, x_n) \leqslant d(x_1, x_2) + ... + d(x_{n-1}, x_n)$$

is indeed true

3.2 Convergence and continuity

Definisjon 2: 3.2.1

Let (X, d) be a metric space. A sequence $\{x_n\}_{n\in\mathbb{N}}$ converges to a point $a \in X$ if there for every $\varepsilon > 0$ exists an $N \in \mathbb{N}$ such that $d(x_n, a) < \varepsilon \forall n \geqslant N$

Lemma 1: 3.2.2

A sequence $\{x_n\}_{n\in\mathbb{N}}$ in a metric space (X, d) converges to a point a, if and only if $d(x_n, 0)$ converges to zero

Bevis 6: 3.2.2

The sequence $d(\{x_n\}, a)$ converges to a point $0 \in X$ if there is for every $\varepsilon > 0 \quad \exists N \in \mathbb{N}$ such that $\forall n \geqslant N$, $d(x_n, a) < \varepsilon$

Can a sequence converge towards multiple points?

Proposisjon 2: 3.2.3

A sequence in a metric cannot converge to more than one point.

Bevis 7: 3.2.3

Assume that $\lim_{n\to\infty} x_n = a$ and $\lim_{n\to\infty} x_n = b$. If we take the triangle inequality on $d(a, b)$, we get

$$d(a, b) \leqslant d(a, x_n) + d(x_n, b)$$

We know that the sequences $d(a, x_n) = d(x_n, a)$ and $d(x_n, b)$ converges to zero if x_n converges to a and b respectively. This

$$\begin{aligned} d(a, b) \leqslant \lim_{n\to\infty} d(x_n, a) + d(x_n, b) \\ = 0 + 0 \end{aligned}$$

And surely, $d(a, b)$ cannot be less than zero. Thus, $d(a, b) = 0 \iff a = b$

Definisjon 3: 3.2.4

Assume (X, d_X) and (Y, d_Y) are two metric spaces. A function $f : X \to Y$ is continuous at a point $a \in X$ if for every $\varepsilon > 0 \exists \delta > 0$ s.t. $d_X(x, a) < \delta \Rightarrow d_Y(f(x), f(a)) < \varepsilon$

Proposisjon 3: 3.2.5

The following is equivalent for a function $f : X \to Y$ for two metric spaces.

- f is continuous at a point $a \in X$
- For all sequences $\{x_n\}$ converging to a, the sequences $\{f(x_n)\}$ are converging to $f(a)$

Bevis 8: 3.2.5

First we show that $i) \Rightarrow ii)$. Assume that f is continuous at a. We want to show that

$$\forall \delta > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geqslant N, d_X(x_n, a) < \delta$$

We know that $\forall \varepsilon > 0 \exists \delta > 0$ s.t. $d_X(x_n, a) < \delta \Rightarrow d_Y(f(x_n), f(a)) < \varepsilon$. Since $x_n \to a$, we know that $\exists N \in \mathbb{N}$ such that $\forall n \geqslant N, d_X(x_n, a) < \delta$. But then $d_Y(f(x_n), f(a)) < \varepsilon$. So $f : X \to Y$ preserves convergence

Second we want to prove that $\neg i) \Rightarrow ii)$ to show a contradiction. Assume that f is *not* continuous at a. Want to show that for all converging sequences $\{x_n\}$ to a, in the metric space X, the sequence $\{f(x_n)\}$ does not converge to $f(a)$ in the metric space Y.

We know that x_n converges to a. This means that for evey $\varepsilon > 0 \exists N \in \mathbb{N}$ such that $d_X(x_n, a) < \varepsilon . \forall n \geqslant N$. Since f is not cont at a,

$$\exists \varepsilon > 0 \text{ such that } \forall \delta > 0, \exists x \in X \text{ such that } d_X(x_n, a) < \delta \Rightarrow d_Y(f(x_n), f(a)) \geqslant \varepsilon$$

Because of the fact that f is not continuous at a, we can pick $\delta = \varepsilon$ and get that $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $d_Y(f(x_n), f(a)) \geqslant \varepsilon \quad \forall n \geqslant N$

Oppgave 5: 3.2.1

Assume that (X, d) is a discrete metric space (recall Example 6 in Section 3.1). Show that the sequence $\{x_n\}$ converges to a if and only if there is an $N \in \mathbb{N}$ such that $x_n = a \forall n \geqslant N$.

Bevis 9: 3.2.1

We have two events

- A: THe sequence $\{x_n\}$ converges to a.
- B: There is an $N \in \mathbb{N}$ such that $x_n = a \forall n \geqslant N$

Our task is to prove $A \iff B$. We begin with $A \Rightarrow B$ Assume first that the sequcence $\{x_n\}$ converges to a. This means that for every $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that $\forall n \geqslant N$, we have that

$$d(x_n, a) < \varepsilon$$

where $d(x_n, a) = \begin{cases} 0, x_n = a \\ 1, x_n \neq a \end{cases}$ We can pick $0 < \varepsilon < 1$ and get that $\forall n \geqslant N$, $d(x_n, a) < 1$, but then $d(x_n, a) = 0$ due to the discrete metric. And since a metric is equal to 0 if and only if $x_n = a$, then there is an $N \in \mathbb{N}$ such that $\forall n \geqslant N$, we have that $x_n = a$

We now prove $B \Rightarrow A$. Since $\exists N \in \mathbb{N}$ such that $x_n = a \quad \forall n \geqslant N$, then $\forall \varepsilon > 0$, we have that $d(x_n, a) = 0 \quad < \varepsilon \forall n \geqslant N$, and therefore, the sequence $\{x_n\}$ converges to a

Oppgave 6: 3.2.2

Prove proposition 3.2.6 without using proposition 3.2.5.

Proposisjon 4: 3.2.6

Let $(X, d_X), (Y, d_Y), (\mathbb{Z}, d_Z)$ be three metric spaces. Assume that

- $f : X \to Y$
- $g : Y \to Z$
- $h : X \to Z$

be the composition $h(x) = g(f(x))$. SHow that if f is continious at $a \in X$, and that g is continious at $b = f(a)$, then h is continuous at a

Bevis 10: 3.2.6

If f is continous at $a \in X$, then $\forall \varepsilon > 0 \quad \exists \delta > 0 s.t. \forall x \in X, d_X(x, a) < \delta \Rightarrow d_Y(f(x), f(a)) < \varepsilon$. If g is continuous at $b = f(a) \in Y$, then $\forall \varepsilon > 0 \quad \exists \delta > 0 s.t. \forall y \in Y, d_Y(y, b) < \delta \Rightarrow d_Z(g(y), g(b)) < \varepsilon$.

This means that $h := g \circ f : X \Rightarrow Z$ is continuous if $\forall \varepsilon > 0 \exists \delta > 0 s.t. \forall x \in X, d_X(x, a) < \delta \Rightarrow d_Z(h(x), h(a)) < \varepsilon$ where $h(a) = g(f(a)) = g(b)$

Oppgave 7: 3.2.5

Let (X, d) be a metric space and choose $a \in X$. Show that $f : X \Rightarrow \mathbb{R}$ given by $f(x) = d(x, a)$ is continuous. Use the metric $d_{\mathbb{R}}(x, y) = |x - y|$

Bevis 11: 3.2.5

f is continious at $a \in X$ if $\forall \varepsilon > 0 \exists \delta > 0 s.t. \forall x \in X, d_X(x, a) < \delta \Rightarrow d_{\mathbb{R}}(f(x), f(a)) < \varepsilon$.

$$d(x, a) = |x - a|$$

$$\begin{aligned} d(f(x), f(a)) &= ||x - a| - |a - a|| \\ &= |x - a| \end{aligned}$$

So we get that f is continuous at a if $\forall \varepsilon > 0 \exists \delta > 0 s.t. \forall x \in X, |x - a| < \delta \Rightarrow |x - a| < \varepsilon$. Just pick $\delta = \varepsilon$, and the proof is complete.

Oppgave 8: 3.2.6

Let $(X, d_X), (Y, d_Y)$ be two metric spaces. A function $f : X \to Y$ is said to be a Lipschitz function if $\exists K \in \mathbb{K}$ such that

$$d_Y(f(u), f(v)) \leqslant K d_X(u, v) \quad \forall u, v \in X$$

Show that all Lipschitz functions are continuous

Bevis 12: 3.2.6

We have that

- A = f is Lipschitz
- B = f is continuous

We are tasked to prove $A \Rightarrow B$. Assume that $f : X \to Y$ is Lipschitz. This means that $\exists K \in \mathbb{R}, K > 0$ such that $d_Y(f(u), f(v)) \leqslant K d_X(y, v) \forall u, v \in X$. A function is continuous at $v \in X$ if $\forall \varepsilon > 0 \exists \delta > 0 s.t. \forall u \in X, d_X(u, v) < \delta \Rightarrow d_Y(f(u), f(v)) < \varepsilon$. We can pick $\delta = \frac{\varepsilon}{K}$ and get that

$$\forall \varepsilon > 0 \exists \delta > 0 s.t. \forall u \in X, d_X(u, v) < \delta = \frac{\varepsilon}{K} \Rightarrow d_Y(f(u), f(v)) \leqslant K d_X(u, v) < K \frac{\varepsilon}{K} = \varepsilon$$

Therefore, f is continous at v.

Oppgave 9: 3.2.8

- a) Assume that $\{x_n\}$ is a sequence in X converging to x. Show that $d(x_n, y) \to d(x, y)$

Bevis 13: 3.2.8 a)

Since $\{x_n\}$ is a sequence converging to $x \in X$, it is equivalent with saying that $d(x, x_n) \to 0$. This is because if $\{x_n\}$ is a sequence converging to $x \in X$, then

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geqslant N, \text{da er } d(x_n, x) < \varepsilon$$

And since we can make ε arbitrarily small, we say that the sequence $\{d(x_n, x)\}$ converges to 0 . To show equivalence the other way, assume that $\{d(x_n, x)\}$ converges to 0. Then

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geqslant N, |d(x_n, x) - 0| = d(x_n, x) < \varepsilon$$

which from the definition of a converging sequence, is equivalent with stating that $x_n \to x$ as $n \to \infty$

From the inverse triangle inequality, we can then state that

$$|d(x_n, y) - d(x, y)| \leqslant d(x_n, x)$$

and since $\{d(x_n, x)\} \to 0$ as $n \to \infty$, then we can pick $\varepsilon := d(x_n, x)$ and get

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geqslant N, |d(x_n, y) - d(x, y)| \leqslant \varepsilon$$

This means that $d(x_n y) \to d(x, y)$ as $n \to \infty$.

- b) Assume that $\{x_n\}$ and $\{y_n\}$ are two sequences in X convergeing respectively to x and y. Show that $d(x_n, y_n) \to d(x, y)$

Bevis 14: 3.2.8 b)

We have from the inverse triangle inequality that

$$\begin{aligned} |d(x_n, y_n) - d(x, y)| &\leqslant |(d(x_n, x) + d(x, y_n)) - (d(x, x_n) + d(x_n, y))| \\ &= |d(x, y_n) - d(x_n, y)| \end{aligned}$$

We have already proved that $d(x_n, y) \to d(x, y)$ and $d(x, y_n) \to d(x, y)$ as $n \to \infty$, so we get that

$$|d(x, y_n) - d(x_n, y)| \to |d(x, y) - d(x, y)| = 0$$

So we have an upper bound

$$|d(x_n, y_n) - d(x, y)| \leqslant 0 \Rightarrow d(x_n y_n) \to d(x, y) \text{ as } n \to \infty$$

3.3 Open and closed sets

Definisjon 4: 3.3.1

Let a be a point in the metric space (X, d) and assume that r is a positive real number. The open ball is the set

$$B(a, r) := \{a \in X : d(a, x) < r\}$$

The closed ball is the set

$$\bar{B}(a, r) := \{a \in X : d(a, x) \leqslant r\}$$

Given a point $x \in X$ where (X, d) is a metric space, there are three intuitive possibilites given a subset $A \subseteq X$

1. There is a ball $B(x, r)$ around x which is contained inside A. In this case, x is an *interior point* in A
2. There is a ball $B(x, r)$ around x which is *not* contained inside A . In other words, its contained inside A^c (all complements are with respect to X, hence $A^c = X\ A$. In this case, x is an *exterior point* of A
3. There is a ball $B(x, r)$ around x which is *both inside and outside* A. So a point $x \in X$ is contained both inside A

and A^c . In this case, x is a boundary point

Definisjon 5: 3.3.2

A subset A of a metric space is *open* if it does not contain any of its boundary points, and A is closed if it contains all of its boundary points

The empty set and the entire metric space X are both open and closed, as they do not have any boundary points.

Oppgave 10: A random number, self made exercise

Prove that the set (c, d) is open, with $c < d$

Bevis 15

Assume that $x \in (c, d)$. The set (c, d) is considered *open* if for all $x \in (c, d)$, there exists an *open* ball centered at x with radius $\varepsilon > 0$. We have that any open ball can be described as

$$B(x, \varepsilon) = \{y \in (c, d) : d(x, y) < \varepsilon\}$$

So if we assume that $x \in (c, d)$, then we can pick $\varepsilon := min\{x - c, x - d\}$, and we are guaranteed to have an open ball.

If we now consider the contrary, we can show that the set $[c, d]$ is not open. $[c, d]$ is not open if $\exists x \in [c, d]$ such that $\forall \varepsilon > 0$, we have that $B(x, \varepsilon) \nsubseteq [c, d]$. Its easy to see that if we pick $\varepsilon := min\{x - c, x - d\}$, then at $x = d$ or $x = c$, we have a radius of 0. But this doesnt satisfy the distance condition of an open ball

$$d(x, y) < \varepsilon$$

And if $\varepsilon = 0$, then $d(x, y) < 0$ which is impossible for a metric. Thus, the set $[c, d]$ is *not* open

Proposisjon 5: 3.3.3

A subset A of a metric space X is open $\iff$ A only consists of interior points. In other words, $\forall a \in A$, there is a ball $B(a, r)$ around a, which is completely contained inside A

Observe that a set A and its complement A^c have exactly the same boundary points. This leads to the following useful proposition

Proposisjon 6: 3.3.4

A subset A of a metric space X is open $\iff$ A^c is closed

Bevis 16: 3.3.4

Assume first that A is open, then A does not contain any of its boundary points, and hence, A^c must be closed.

Conversely, assume that A^c is closed, then it contains all of its boundary points, and thus, the boundary points of A^c cant be contained inside A and A must as a result be open.

We can extract the boundary points and interior points of an arbitrary set A

- We define
$$int(A) := \{x : \text{x is an interior point}\}$$
- and the closure of A as
$$cl(A) := \{x : x \in A \text{ or x is a boundary point of A}\}$$

To denote the boundary of a set A, we write

- $\partial A = \{\text{all boundary pts of A}\}$

We write the closure of a set A as $cl(A) = \bar{A} = A \cup \partial A$

We know that a set A is open if it can be written as

- $A = int(A)$
- and A is open if the following statement is true; $A \cap \partial A = \varnothing$

Furthermore, we know if a set A is closed if

- $A = int(A)$
- $\partial A \subseteq A$

Proposisjon 7: 3.3.5

$int(A)$ is an open set, and $cl(A)$ is a closed set

Bevis 17: 3.3.5

Lemma 2: 3.3.6

All open balls $B(a,r)$ are open sets and all closed balls $\bar{B}(a,r)$ are closed sets

Bevis 18: 3.3.6

Assume that $x \in B(a,r)$. We need to show that $\exists B(x,\varepsilon) \subseteq B(a,r)$. If we choose

$$\varepsilon := r - d(x,a)$$

then we see that if $y \in B(x,\varepsilon)$, then by the triangle inequality that

$$d(y,a) \leqslant d(y,x) + d(x,a)$$

Now remember that since $y \in B(x,\varepsilon)$, we know that

$$\begin{aligned} d(y,a) \leqslant d(y,x) + d(x,a) < \varepsilon + d(x,a) \\ &= r - d(x,a) + d(x,a) \\ &= r \end{aligned}$$

thus we have proved that $d(y,a) < r$, which means that $B(x,\varepsilon) \subseteq B(a,r)$. We come to this conclusion since $y \in B(x,\varepsilon)$

Now we prove that a closed ball is a closed set. There are multiple ways to approach this. We can use the fact that if $\{x_n\}$ is a convergent sequence in $\bar{B}(a,r)$, then it must converge towards $x \in \bar{B}(a,r)$. Another proof is to show that the set $X \backslash \bar{B}(a,r)$ is an open set.

We define $F = \bar{B}(a,r)$ which is the closed ball. So we need to show that F^c is open. A set is open if any $y \in F^c$ has the property $d(y,a) > r$. Since $y \notin F$, then $d(y,a) > r$. And from the triangle inequality, we get that

$$d(y,a) \leqslant d(y,x) + d(x,a)$$

We now define

$$B(y,\varepsilon) = \{x \in X : d(y,x) < \varepsilon\}$$

and claim that $B(y,\varepsilon) \subseteq F^c$ where $\varepsilon := d(y,a) - r$. let $x \in B(y,\varepsilon)$ and get that $d(x,y) < \varepsilon$. Then

$$d(y,a) \leqslant d(y,x) + d(x,a) \Rightarrow d(y,a) < \varepsilon + d(x,a)$$

So

$$d(y,a) - \varepsilon < d(x,a) \Rightarrow d(y,a) - d(y,a) + r < d(x,a) \Rightarrow r < d(x,a)$$

Which proves that $x \notin F$. Therefore, $x \in F^c$ and our claim that $B(y,\varepsilon) \in F^c$ is therefore true

Proposisjon 8: 3.3.7

Assume that F is a subset of a metric space X, then TFAE

- F is closed
- $\forall$ convergent sequences $\{x_n\}_{n\in\mathbb{N}}$ of elememts in F, the limit $a = \lim_{n\to\infty} x_n$ also belongs to F

Bevis 19: 3.3.7

Assume that F is closed, and that a is not in F . Since F is closed, it contains all of its boundary points. Which means that a has to be an exterior point. And hence there is a ball around a that only has elements inside F^c, but then, $\{x_n\}$ can never converge to a, and we have a contradiction

Assume now that F is not closed.

Proposisjon 9: 3.3.9

Let $f : X \to Y$ be a function between metric spaces, and let $x_0 \in X$. Then the following are equivalent

- f is continious at x_0
- For all neighbourhoods V of $f(x_0)$, there is a neighbourhood U of x_0 such that $f(U) \subseteq V$

Proposisjon 10: 3.3.10

The following are equivalent for a function $f : X \to Y$ between two metric spaces.

- f is continuous
- Whenever V is an open subset of Y, the inverse image $f^{-1}(V)$ is an open set in X

Proposisjon 11: 3.3.11

The following are equivalent for a function $f : X \to Y$ between two metric spaces.

- f is continuous
- Whenever F is an closed subset of Y, the inverse image $f^{-1}(F)$ is an closed set in X

Proposisjon 12: 3.3.12

Let (X, d) be a metric space

1. If $\mathcal{G}$ is a finite or infinite collection of open sets, then the union

$$\cup_{G \in \mathcal{G}} G$$

   is open

2. If $G_1, G_2, ..., G_n$ is a finite collection of open sets, then the intersection

$$G_1 \cap G_2 \cap ... \cap G_n$$

   is open

Proposisjon 13: 3.3.13

Let (X, d) be a metric space

1. If $\mathcal{F}$ is a finite or infinite collection of closed sets, then the intersection $\cap_{F \in \mathcal{F}} F$ is closed
2. If $F_1, F_2, ..., F_n$ is a finite collection of closed sets, then the intersection

$$F_1 \cup F_2 \cup ... \cup F_n$$

   is closed

Oppgave 11: 3.1.1

Consider $X = \mathbb{R}$ with the canonical metric $d(x, y) = |x - y|$. SHow that the interval (a, b) is open, and that $[a, b]$ is closed for any $a, b \in \mathbb{R}$ with $a \leqslant b$. What happens when $a = b$? Explain why $(a, b]$ is neither open nor closed when $a < b$

Bevis 20: 3.3.1

First task is to show that the interval (a, b) is open. To prove this, we show that the open ball centered at $x \in (a, b)$ defined by

$$B(x, r) = \{y \in \mathbb{R} : |x - y| < r\}$$

is contained within (a, b)

We define

$$r := |a - b| - \left|x - \frac{a+b}{2}\right|$$

then if $y \in B(x, r)$, we have that

$$\left|y - \frac{a+b}{2}\right| \leqslant |y - x| + \left|x - \frac{a+b}{2}\right|$$

But since $y \in B(x, r)$ we have that

$$|x - y| < r = |a - b| + \left|x - \frac{a+b}{2}\right|$$

and so

$$\left|y - \frac{a+b}{2}\right| < |a - b| - \left|x - \frac{a+b}{2}\right| + \left|x - \frac{a+b}{2}\right|$$

and so

$$\left|y - \frac{a+b}{2}\right| < |a - b|$$

which means that $y \in (a, b)$. Thus the ball $B(x, r) \subseteq (a, b)$

Now we prove that $[a, b]$ is a closed set. We look at the complement and observe that $(-\infty, a) \cup (b, \infty)$ is an open set. So for a $x \in [a, b]^c$, we need to prove that $B(x, r) \subseteq [a, b]^c$

Assume $y \in B(x, r)$, then $|y - m| > |m - a|$ where $m := \frac{a+b}{2}$. Now define

$$r := |x - m| - |a - m|$$

From the inverse triangle inequality, we get

$$|x - m| \leqslant |x - y| + |y - m|$$

But since $y \in B(x, r)$, we have that $|x - y| < r$ so

$$\begin{aligned} |x - m| &< r + |y - m| \\ |x - m| - r &< |y - m| \\ |x - m| - |x - m| + |a - m| &< |y - m| \\ |a - m| &< |y - m| \end{aligned}$$

so we get that $y \notin [a, b]$, so $y \in [a, b]^c$. Thus, since $y \in B(x, r)$, we conclude that $B(x, r) \subseteq [a, b]^c$. We have now proved that the complement of $[a, b]$ is open, because it contains every open circle with center inside of it. I.e. it contains all interior points and is thus open. From this, we can conclude the $[a, b]$ is a closed set.

Now consider the case $[a, b]$ when $a = b$. In this case, we have the empty set, which doesnt have any boundary points, but also doesnt contain any points. Thus it is both open and closed.

Furthermore, we have that $A = (a, b]$ is neither open nor closed. Its not open since it contains the boundary point b, and its closed if it contains its interior points as well as boundary points. Since it does not contain the boundary point a, its not closed. If A is not open, then

$$\exists x \in (a, b] \text{ such that } \forall \varepsilon > 0, B(x, \varepsilon) \not\subset (a, b]$$

We know that b is a part of the set A. If we pick for example $B(b, \varepsilon)$, then $\exists y \in B(b, \varepsilon)$ where $y \notin A$, thus A is not open.

To show that A is not closed, assume that A is closed. Then A^c is open.

$$A^c = (\infty, a] \cup (b, \infty)$$

This set is not open, since we can select x close to a and make ε such that $B(x, \varepsilon)$ contains elements not in A^c

Oppgave 12: 3.3.1

Assume that (X, d) is a discrete metric space

- a) Show that an open ball in X is either a set with only one element, a singleton, or all of X
- b) Show that all subsets of X are both open and closed
- c) Assume that (Y, d_Y) is another metric space. Show that all functions $f : X \to Y$ are continuous.

Bevis 21: 3.3.1 a)

We have that

$$d(x, y) = \begin{cases} 0, x = y \\ 1, x \neq y \end{cases}$$

Any open ball in X is such that

$$\forall x \in X \exists \varepsilon > 0 \text{ such that } B(x, \varepsilon) = \{y \in X : d(x, y) < \varepsilon\}$$

Assume now that $y \in X$, then $d(x, y) < \varepsilon$. We can pick ε such that

$$\varepsilon < 1$$

But then $d(x, y) < 1$. This means that the ball $B(x, \varepsilon) = \{y \in X : d(x, y) = 0\}$ which is only one unique point. Therefore, the ball is a singleton

We can now pick $\varepsilon > 1$ and get that $d(x, y) < \varepsilon \forall x, y \in X$. This means that the ball $B(x, \varepsilon)$ contains all of X.

Bevis 22: 3.3.1 b)

Assume that $A \subseteq X$. A is open if

$$\forall x \in A \exists \varepsilon > 0 \text{ such that } B(x, \varepsilon) \subseteq A$$

Since A can either be a singleton or all of X, it is easy to see that such an $\varepsilon > 0$ exists given $x \in A$. Consider when A is a singleton. Then all points in A is the point where $x = y$. In this case, any $\varepsilon > 0$ will give us $d(x, y) < \varepsilon$. This means that any $y \in B(x, \varepsilon)$ also has the propery that $y \in A$. Consider now the case when A is all of X. We can pick $\varepsilon < 1$ and get that all points inside $y \in B(x, \varepsilon)$ also has the property that $y \in A$ and thus, A is open.

Now to show that A is closed, we know that the following are equivalent

- A is closed
- $\forall \{x_n\} \in X$ converging to $x \in X$, then $x \in A$

We have proved from section 3.2 that

$$d(x_n, y) \to d(x, y) \text{ as } n \to \infty$$

. Assume that (x_n) converges to $x \in X$. This is equivalent with saying that $d(x_n, y) \to d(x, y)$ as $n \to \infty$. We need to show that $x \in A$.

Since $d(x, y) < \varepsilon$, we can pick ε to be bigger than or less than 1, depending on if A is a singleton or all of X. If A is a singleton, then any ε will make $x \in A$. Same for if A is all of X, then we can pick $\varepsilon > 1$ and get that for all converging sequences $(x_n) \in X$ converging to $x \in X$, then $x \in A$, which is equivalent with saying that A is closed.

Bevis 23: 3.3.1 b)

Assume that $A \subseteq X$ where (X, d) is the discrete metric space given by

$$d(x,y) = \begin{cases} 0, x = y \\ 1, x \neq y \end{cases}$$

Lets first prove that A is an open subset. A set is open if for any $x \in A$, there exists $\varepsilon > 0$ such that the ball $B(x, \varepsilon) \subseteq A$. Assume that $x \in A$. We can pick $\varepsilon = \frac{1}{2}$ and see that for the ball

$$B\left(x, \frac{1}{2}\right) = \left\{y \in X : d(x,y) < \frac{1}{2}\right\}$$

Pick any $y \in X$, then y is in A if $d(x,y) < \frac{1}{2}$, but this implies that $d(x,y) = 0$ as we are in a discrete metric space. This means that $x = y$, and $B(x, \varepsilon) = \{x\} \subseteq A$ because we assumed that $x \in A$

To prove that A is a closed set, we prove that A^c is open. Assume that $x \in A^c$, then the ball centered at $B\left(x, \varepsilon \frac{1}{2}\right) = \{x\} \subseteq A^c$

Bevis 24: 3.3.1 c)

To prove that any $f : X \to Y$ if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } \forall x \in X, d(x,y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon$$

We have that (Y, d_Y) is another general metric space. We have that

$$d(x,y) = \begin{cases} 0, x = y \\ 1, x \neq y \end{cases}$$

So we can pick $\delta = 1$ and get that when $d(x,y) < \delta$, we have that $d(x,y) < 1 \Rightarrow d(x,y) = 0 \iff x = y$. So in this case, $d(f(x), f(y)) = d(f(x), f(x)) < \varepsilon$. Thus, f satisfies the continuity definition, and any $f : X \to Y$ is continuous.

Bevis 25: 3.3.1 c)

An alternative is to use proposition 3.3.9. Assume that $x_0 \in X$. Then any ball $B(x_0, 1) = \{x_0\}$. From proposition 3.3.9, we have that

$$\forall \text{ neighbourhoods V of } f(x_0), \exists U \text{ of } x_0 \text{ such that } f(x_0) \subseteq V.$$

Take now $U \subseteq X$ and $V \subseteq Y$, then U is open and any ball $B(x_0, 1) = \{x_0\} = U$. This means that

$$\{f(x_0)\} \subseteq V$$

and this is equivalent with f being continuous at x_0.

Oppgave 13: 3.3.2

Give a geometric description of the ball $B(a, r)$ in the Manhattan metric. Make a drawing of a typical ball. Show that the Manhattan metric and the usual metric in $\mathbb{R}^2$ have exactly the same open sets.

Bevis 26: 3.3.2

Oppgave 14: 3.3.3

Assume that F is a nonempty, closed and bounded subset of R (with the usual metric $d(x,y) = |y - x|$). Show that $supF \in F$ and $infF \in F$. Give an example of a bounded, but not closed set F such that $supF \in F$ and $infF \in F$.

Oppgave 15: 3.3.7

Assume $A \subseteq (X, d)$. Show that interior pts of A are exterior pts of A^c, and that interior pts of A^c are exterior pts of A. Show also that the boundary pts of A are the boundary pts of A^c

Bevis 27: 3.3.7

Assume that $x \in A$. x is an interior point if any ball centered at x is contained within A. So

$$B(x, \varepsilon) = \{y \in A : d(x, y) < \varepsilon\} \subseteq A$$

From definition, a point x' is an exterior point if there exists a ball $B(x', \varepsilon) \nsubseteq A$. To show that an interior point of A is an exterior point of A^c, we know that since $B(x, \varepsilon) \subseteq A$, then $B(x, \varepsilon)$ is not contained inside A^c, and is thus an exterior point of A^c

Now we show that an interior point of A^c is an exterior point of A Assume that $x \in A^c$. x is an interior point of A^c if any open ball centered at an interior point of A^c is contained within A^c.

$$B(x, \varepsilon) = \{y \in A^c : d(x, y) < \varepsilon\} \subseteq A^c$$

Since $B(x, \varepsilon) \subseteq A^c$, we know that $x \notin A$, and is therefore an exterior point of A

Now we show that the boundary points of A are the boundary points of A^c. Assume that $x, y \in B(z, \varepsilon)$ where $z \in A$ and $x \in A, y \in A^c$. A point in A is a boundary point if the centered at that point contains both elements in A and A^c. Since $x \in A$ and $y \in A^c$, we know that z is a boundary point. And since the ball centered at z contains both elements inside A and inside A^c, we know that $z \in A$ and $z \in A^c$

Oppgave 16: 3.3.11

Prove proposition 3.3.12. Find an example of an infinite collection of sets $G_1, G_2, ...$ whose intersection is not open.

Bevis 28

- a) We have that $G_1, G_2, ...G_n, ...$ are all open sets. Assume that x is inside one of $G_1, G_2, ...G_n, ...$, then $x \in \cup_{G \in \mathcal{G}} G$ where $\mathcal{G}$ is the family of sets of all Gs. Since x is inside at least one of these sets, and all these sets are open, then $\cup_{G \in \mathcal{G}} G$ is an open set.
- b) If $G_1, ..., G_n$ is a finite collection of open sets, then

  $$G_1 \cap ... \cap G_n = \{g : \text{g belongs to all} G_k \forall k \in K\}$$

  where K is some index set. If we assume $g \in G_k \forall k \in K$, then $g \in \cap_{k=1}^{n} G_K$, which means that g belongs to all G. Since all $G_k \forall k \in K$ are open, $\cap_{k=1}^{n} G_K$ is also open

Oppgave 17: section 3.3

Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$ be a continuous function, let $y \in Y$ be some given point, and consider the problem of finding an $x \in X$ such that $f(x) = y$. Prove that the set of solutions of this equation is a closed set.

Bevis 29

If we assume that $y \in Y$ is given to us as a point to the equation $f(x) = y$, then the singleton $\{y\}$ is a closed subset where $\{y\} \subseteq Y$. From prop. 3.3.11, we have that since f is a continuous function, then $f^{-1}(\{y\})$ is a closed set in X.

A singleton is a set that contains a single element. So

$$\{1, 2, 3, 4\}$$

are not a singleton set. But

$$\{\{1\}, \{2\}, \{3\}, \{4\}\}$$

is a set of singletons. Also, if $x \in \{1, 2, 3, 4\}$, its the same as saying that

$$x \in \{1\} \cup \{2\} \cup \{3\} \cup \{4\} = \cup_{k=1}^{n} \{i\}$$

Oppgave 18: section 3.3

Consider $X = \mathbb{R}$ with canonical metric $d(x, y) = |x - y|$. The support of a function $f : \mathbb{R} \to \mathbb{R}$ is the set of points x where $f(x) \neq 0$. The support of a function is defined to be the closure of the above set, which is of course closed, not open.

Bevis 30

Assume that $x \in supp(f) = \{x \in \mathbb{R} : f(x) \neq 0\}$. Then $f(x) \neq 0$ and $d(f(x), 0) = |f(x)| = |f(x)|$. Since f is continuous, we have that $\forall y \in \mathbb{R}$

$$|f(x) - f(y)| < \varepsilon$$

for some $\varepsilon > 0$ when

$$|x - y| < \delta$$

By the inverse triangle inequality, we have that

$$d(f(x), f(y)) = |f(x) - f(y)| \leqslant |f(x)| + |f(y)|$$

which means that by the inverse triangle inequality,

$$||f(x) - f(y)| - |f(y)|| \geqslant |f(x)| > 0$$

So any neighborhood around $x \in supp(f)$ is an open ball inside $supp(f)$, proving that $supp(f)$ is an open set.

Bevis 31: Alternative to support of f is open

The above proof is incorrect.

We can instead view the support of f as the set

$$supp(f) = \left\{x \in \mathbb{R} : x \neq f^{-1}(0)\right\} = f^{-1}\left(\mathbb{R} \backslash \{0\}\right)$$

since $\mathbb{R}$ is an open set (we can place any open circle inside it, and still have it a subset of $\mathbb{R}$), and we are subtracting finite singletons, we get that $supp(f)$ is an open set.

Bevis 32: Another alternative to support of f being open

If we define

$$V := \mathbb{R} \backslash f^{-1}\left(\{0\}\right) = supp(f)$$

Now we can just show that V is open. V is open because $V = (-\infty, 0) \cup (0, \infty)$ is the union of two open sets, and is therefore open in itself. Since f is continuous, and V is an open subset of $\mathbb{R}$, then the inverse image $f^{-1}(V)$ is an open subset in $\mathbb{R}$. And the proof is concluded.

3.4 Complete Spaces

The main reason that calculus in $\mathbb{R}$ and $\mathbb{R}^n$ is so successfull, is because $\mathbb{R}$ is complete. We will now generalize this notion of completeness to metric spaces.

Definisjon 6: 3.4.1

A sequence $\{x_n\}$ in a metric space (X, d) is a cauchy sequence if $\forall \varepsilon > 0 \, \exists N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon \, \forall n, m \geqslant N$

Proposisjon 14: 3.4.2

Every convergent sequence is a Cauchy sequence

Bevis 33: 3.4.2

Assume that $\{x_n\}$ is a convergenet sequence to $x \in X$. Then $\forall \varepsilon > 0 \, \exists N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon \, \forall n \geqslant N$. But then we can just pick another $m \geqslant N$ and get that $d(x_m, x) < \varepsilon \, \forall m \geqslant N$. Then from the triangle inequality, we get that

$$d(x_n, x_m) \leqslant d(x_n, x) + d(x_m, x) < \varepsilon + \varepsilon = 2\varepsilon$$

But then $\{x_n\}$ is a cauchy sequence, since we can pick two indeces above N and they both satisfy

$$d(x_n, x_m) < \varepsilon \, \forall n, m \geqslant N$$

Definisjon 7: 3.4.3

A metric space is called *complete* if all cauchy sequences converge.

As an example, we know that all cauchy sequences in $\mathbb{R}^n$ converge, but that $\mathbb{Q}$ does not

The reason we care about complete metric spaces, is that we often want to approximate solutions. When our approximation converges to the right solution, we typically need a complete metric space to avoid the "holes"in the metric space

Proposisjon 15: 3.4.4

Assume (X, d) is a complete metric space. If A is a subset of X, then (A, d_A) is complete if and only if A is closed

Bevis 34: 3.4.4

Assume first A is closed. If $\{a_n\}$ is a cauchy sequence in A. $\{a_n\}$ is also a cauchy sequence in X, and since X is complete, $\{a_n\}$ converges to a point $a \in X$. SInce A is closed, proposition 3.3.7 tells us that $a \in A$. But then $\{a_n\}$ converges to $a \in (A, d_A)$, and hence, (A, d_A) is complete.

If A is not closed, there is a boundary point a that does not belong to A. Each ball $B\left(a, \frac{1}{n}\right)$ must contain an element a_n from A. In X, the sequence $\{a_n\}$ converges to a, and must be a cauchy sequence. However, since $a \notin A$, the sequence $\{a_n\}$ does not converge to a point in A. Hence, we have found a cauchy sequence in (A, d_A) that does not converge to a point in A, hence (A, d_A) is incomplete.

We now want to find solutions by repeated approximation, and we use the completion property to help us. If we start with the initial point x_0, then we define a function $f : X \to X$ such that

$$x_{k+1} = f(x_k)$$

This can be viewed as an evolving system. This system has reached equilibrium if x_{k+1} is a fixed point. That is if

$$x_k = f(x_k)$$

To reformulate, we need a few more definitons. A function $f : X \to X$ is called a *contraction* if there is a positive number $s < 1$ such that

$$d\left(f(x), f(y)\right) \leqslant s d(x, y) \; \forall x, y \in X$$

We call s a contraction factor for f. I will prove later that all contractions are continuous (Lipschitz continuity). We get that by induction,

$$d\left(f^{\circ n}(x), f^{\circ n}(y)\right) \leqslant s^n d(x, y)$$

where $f^{\circ n}(x) = f\left(f\left(f\left(...f(x)...\right)\right)\right)$ is the result of iterating f exactly n times.

Teorem 1: 3.4.5 Banachs Fixed Point theorem

Assume (X, d) is a complete metric space and that $f : X \to X$ is a contration. Then f has a unique fixed point a, and no matter which starting point $x_0 \in X$ we choose, the sequence

$$x_0, x_1 = f(x_0), x_2 = f^{\circ 2}(x_0), ..., x_n = f^{\circ n}(x_0)$$

converges to a

Bevis 35: 3.4.5

Let $x_0 \in X$. Define $x_1, x_2, ..., \in X$ as

$$x_{n+1} = f(x_n)$$
$$x_n = f^{\circ n}(x_0)$$

Let $n \in \mathbb{N}$, then

$$d\left(x_n, x_{n+1}\right) = d\left(f(x_{n-1}), f(x_n)\right) \leqslant s\dot{d}\left(x_{n-1}, x_n\right) \leqslant ... \leqslant s^n d\left(x_0, x_1\right)$$

Next, for $n \in \mathbb{N}$ and $d \in \mathbb{N}$, then we get from the triangle inequality that

$$d(x_n, x_{n+d}) \leqslant \sum_{k=0}^{d-1} d\left(x_{n+k}, x_{n+k+1}\right)$$

$$d\left(x_n, x_{n+d}\right) \leqslant \sum_{k=0}^{d-1} d\left(x_{n+k}, x_{n+k+1}\right) \leqslant \sum_{k=0}^{d-1} s^{n+k} d\left(x_0, x_1\right) = d\left(x_0, x_1\right) s^n \frac{1}{1-s}$$

The last step, comes from the fact that

$$\sum_{k=0}^{\infty} s^k = \frac{1}{1-s}$$

since $s \in [0, 1]$. Hence, if $n, m \geqslant N$, then

$$d\left(x_n, x_m\right) \leqslant \frac{d\left(x_0, x_1\right)}{1-s} \dot{s}^N \to 0 \text{ as } N \to \infty$$

So $\{x_n\}$ is cauchy. Hence $x_n \to x$ as $N \to \infty$

What about uniqueness?

x is a fixed point since,

$$x_{n+1} = f(x_n)\ \forall n$$

$$x = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right) = f(x)$$

Let x and y be fixed points.

$$x = f(x),\ y = f(y)$$

$$d(x, y) = d\left(f(x), f(y)\right) \leqslant sd\left(x, y\right)$$

so either $d\left(x, y\right) = 0$ or $1 \leqslant s$. The latter is impossible, so $d\left(x, y = 0\right) \iff x = y$, hence uniqeness is proved.

Oppgave 19: section 3.4

Prove that $(\mathbb{Z}, d)$ where $d\left(x, y\right) = |x - y|$

Bevis 36

A metric space is called complete if all cauchy sequences converge. All convergent sequences are convergent. Take any $\{x_n\} \in \mathbb{Z}$. I must prove that any $\{x_n\} \in \mathbb{Z}$ is convergent. $\{x_n\}$ is convergent to $x \in \mathbb{Z}$ if $\forall \varepsilon > 0 \exists N \in \mathbb{N}$ such that

$$|x - y| < \varepsilon\ \forall n \geqslant N$$

Pick $\varepsilon \in (0, 1)$, then $|x_n - x| < \varepsilon$ implies that $x = y$, since $x, y \in \mathbb{Z}$. Thus, $\exists N \in \mathbb{N}$ such that $\forall \varepsilon \in (0, 1)$ $|x_n - x| < \varepsilon\ \forall n \geqslant N$ that implies that $x_n = x\ \forall n \geqslant N$. Hence $\{x_n\}$ is convergent, this also a cauchy sequence. Since all convergent sequences in $\mathbb{Z}$ are cauchy sequences, $(\mathbb{Z}, d)$ is a complete metric space.

Bevis 37: Alternative proof section 3.4

The first proof uses the implication that *convergent* $\Rightarrow$ *Cauchy*. According to chatgpt, its not enough. Must use that *Cauchy* $\Rightarrow$ *convergent*. Why this matters, is because every convergent sequence in $\mathbb{Q}$ is cauchy. Yet $\mathbb{Q}$ is not complete, because there exists cauchy sequences (for example approximations to $\sqrt{2}$) that do not converge in $\mathbb{Q}$

Instead, take any cauchy sequence inside $(\mathbb{Z}, d)$. It is convergent if $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $|x_n - x_m| < \varepsilon\ \forall n, m \geqslant N$. Pick now any $\varepsilon \in (0, 1)$, then since x_n and $x_m \in \mathbb{Z}$, then $|x_n - x_m| < \varepsilon$ implies that $x_n = x_m\ \forall n, m \geqslant N$

Since any cauchy sequence converges to a point, we have that the metric space $(\mathbb{Z}, d)$ is complete.

Oppgave 20: 3.4.2

Assume that (X, d_X) and $(Y d_Y)$ are complete spaces, and give $X \times Y$ the metric

$$d\left(\left(x_1, x_2\right), \left(y_1, y_2\right)\right) = d_X\left(x_1, x_2\right) + d_Y\left(y_1, y_2\right)$$

Show that $(X \times Y, d)$ is complete.

Bevis 38: 3.4.2

Since (X, d_X) and (Y, d_Y) are complete metric spaces, all cauchy sequences converge.

This means that for a cauchy sequence $\{x_{n_1}\}$ in (X, d_X), we have $\forall \varepsilon_1 > 0\ \exists N_1 \in \mathbb{N}$ such that $d_X(x_{n_1}, x_{m_1}) < \varepsilon_1\ \forall n_1, m_1 \geqslant N_1$

And that for a cauchy sequence $\{y_{n_2}\}$ in (Y, d_Y), we have $\forall \varepsilon_2 > 0\ \exists N_2 \in \mathbb{N}$ such that $d_Y\left(y_{n_2}, y_{m_2}\right) < \varepsilon_2\ \forall n_2, m_2 \geqslant N_2$

The new metric space $(X \times Y, d)$ with metric

$$d\left(\left(x_1, x_2\right), \left(y_1, y_2\right)\right) = d_X\left(x_1, x_2\right) + d_Y\left(y_1, y_2\right)$$

is complete if $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that

$$d\left(\left(x_n, x_m\right), \left(y_n, y_m\right)\right) < \varepsilon\ \forall n, m \geqslant N$$

If we now pick $\varepsilon := \varepsilon_1 + \varepsilon_2$, we get that $\forall n_1, m_1, n_2, m_2 \geqslant N$

$$d\left(\left(x_{n_1}, x_{m_1}\right), \left(y_{n_2}, y_{m_2}\right)\right) = d_X\left(x_{n_1}, x_{m_1}\right) + d_Y\left(y_{n_2}, y_{m_2}\right) < \varepsilon_1 + \varepsilon_2 = \varepsilon$$

Thus any cauchy sequence in $X \times Y$ is convergent, hence proves that $(X \times Y, d)$ is a complete metric space.

Oppgave 21: 3.4.4

Assume that d_1 and d_2 are two metrics on the same space X. We say that d_1 and d_2 are equivalent if there are constants K and M such that $d_1(x, y) \leqslant K d_2(x, y)$ and $d_2(x, y) \leqslant M d_1(x, y)\ \forall x, y \in X$. Show that if d_1 and d_2 are equivalent, and one of the spaces $(X, d_1), (X, d_2)$ are complete, then so is the other.

Bevis 39: 3.4.4

Assume that (X, d_1) is complete. That is, for some $\{x_n\} \in (X, d_1)$ we have that $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $d_1(x_n, x_m) < \varepsilon\ \forall n, m \geqslant N$. But since d_1 and d_2 are equivalent, we have that for some constant M that

$$d_2(x, y) \leqslant M d_1(x, y)$$

and hence, we also have that $d_2(x_n, x_m) < \varepsilon\ \forall n, m \geqslant N$. This means that (X, d_2) is also complete when (X, d_1) is complete.

Assume the opposite, that (X, d_2) is complete. That is, for some $\{x_n\} \in (X, d_2)$, we have that $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $d_2(x_n, x_m) < \varepsilon\ \forall n, m \geqslant N$. But since d_1 and d_2 are equivalent, we have that for some constant K that

$$d_1(x, y) \leqslant K d_2(x, y)$$

and hence, we also have that $d_1(x_n, x_m) < \varepsilon\ \forall n, m \geqslant N$. This means that (X, d_1) is also complete when (X, d_2) is complete.

This exercise a different metric doesnt necessarily change the complete property of the metric space, as long as the metrics are equivalent.

Oppgave 22: 3.4.5

Assume that $f : [0, 1] \to [0, 1]$ is a differentiable function and that there is a number $s < 1$ such that $|f'(x)| < s\ \forall x \in (0, 1)$. Show that there is exactly one point $a \in [0, 1]$ such that $f(a) = a$

Bevis 40: 3.4.5

The first task is to show that f is a contraction, that is $\exists s < 1$ such that $d(f(x), f(y)) \leqslant s d(x, y)\ \ \forall x, y \in X$.

Consider now the metric space $([0, 1], d(x, y) = |x - y|)$. Since f is differentiable, we have that

$$f(x) = \int_0^x f'(\theta)\, d\theta - f(0)$$

So

$$|f(x) - f(y)| = \left| \int_0^x f'(\theta)\, d\theta - \int_0^y f'(\varphi)\, d\varphi \right|$$

We assume that $\exists s < 1$ such that $|f'(x)| < s\ \forall x \in [0, 1]$. So

$$|f(x) - f(y)| = \left| \int_y^x f'(t) dt \right| \leqslant \int_y^x \left| f'(t) \right| dt < s \int_y^x dt = s\,|x - y|$$

Derfor er f en kontraksjon. Må nå bruke Banachs fikspunkt teorem å vise at det finnes en unik $a \in (0, 1)$ slik at $f(a) = a$. Dette følger direkte fordi f er en kontraksjon, i tillegg til at $[0, 1]$ er et komplett metrisk rom (fordi det er lukket)

Bevis 41: 3.4.5 alternative

With the mean value theorem, we have a $c \in [0, 1]$ such that

$$f'(c) = \frac{f(x) - f(y)}{x - y} \Rightarrow \left| f'(c) \right| = \frac{|f(x) - f(y)|}{|x - y|} \Rightarrow |f(x) - f(y)| = \left| f'(c) \right| \cdot |x - y|$$

This means that

$$d(f(x), f(y)) = \left| f'(c) \right| \cdot d(x, y) < s d(x, y)$$

Oppgave 23: 3.4.6

You are standing with a map in your hand inside the area depicted on the map. Explain that there is exactly one point on the map that is vertically above the point it depicts.

Bevis 42: 3.4.6

If we are standing with a map of the area we are standing in, then we can think of $f : X \to X$ as a contraction, where $X \subseteq \mathbb{R}^2$ is the map, and $f(x), x \in X$ is the depicted point on the ground. So take two points $x, y \in X$, then their distance on the ground is $d(x, y)$. On the map, the distance between the depicted points are a lot shorter than the points on the ground. Thus, $f(x)$ represents the depicted point on the map. Thus, we scale down the number $d(x, y)$ by a constant $s \in [0, 1)$, and get that $f : X \to X$ is a contraction

Oppgave 24: 3.4.8

A subset D of a metric space X is dense if for all $x \in X$ and all $\varepsilon \in \mathbb{R}_+$ there is an element $y \in D$ such that $d(x, y) < \varepsilon$. Show that if all Cauchy sequences $\{y_n\}$ from a dense set D converge in X , then X is complete.

Bevis 43: 3.4.8

Take any $\{y_n\}$ cauchy sequence in D. Assume D is dense. $\{y_n\}$ is a convergent cauchy sequence if $\forall y_n \in D$, $\exists y_m \in X$ such that $d(y_n, y_m) < \varepsilon$. Since $\{y_m\}$ is another sequence in D, we have that

3.5 Compact sets

Assume we have a sequence $(x_n)_{n\in\mathbb{N}}$. Then we can define the subsequences

$$\{y_k\} = \{x_{n_k}\}$$

where $\{x_{n_k}\}$ is a subsequence. If we consider the indeces $n_1 < n_2 < ... < n_k < ...$ of the sequence. Then any subsequence will naturally just be the elements of the original sequences, but a few items dropped

Proposisjon 16: 3.5.1

If the sequence $\{x_n\}$ converges to a , then so do all subsequences

Bevis 44: 3.5.1

Assume that $\{x_n\}$ converges to a, then $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $d(x_n, a) < \varepsilon\ \forall n \geqslant N$.

$$x_n = (x_1, x_2, ..., x_n, ...)$$

if we take a subsequence of x_n we get (x_{n_k}) where $n_1 < n_2 < ... < n_k < ....$ Since n_k is an increasing sequence, there exists K such that $n_K \geqslant N\ \forall k \geqslant K$. But since $n \geqslant N$, we have that $d(x_{n_k}, a) < 0\ \forall n_k \geqslant N$. Thus (x_{n_k}) is also converging towards a. We get this justification because (n_k) is an increasing sequence.

Take for instance the sequence

$$x_n = (0, 1, 0, 1, ..., 0, 1, 0, ...)$$

And pick the subsequence

$$x_{n_k} = (0, 0, 0, 0, ...)$$

which is every other element. Then obviously, we have that $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\forall n \geqslant N\ d(x_{n_k}, 0) < \varepsilon$.

If we pick x_n to already be a converging sequence, for. example

$$x_n = \frac{1}{n}$$

then $n = (1, 2, 3, 4, ...)$ and $n_k = (3, 5, 7, 9, ...)$ then of course, $x_{n_k} = \left(\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, ...\right)$ also converges to 0, just as x_n does

Definisjon 8: 3.5.2

A subset K of a metric space (X, d) is called a compact set if every sequence in K has a subsequence converging to a point in K. The space (X, d) is compact if X is a compact set. i.e. if all sequences in X have a convergent subsequence

Definisjon 9: 3.5.3

A subset A of a metric space (X, d) is bounded if there is a number $M \in \mathbb{R}$ such that $d(a, b) \leqslant M \ \forall a, b \in A$

Proposisjon 17: 3.5.4

Every compact set K in a metric space (X, d) is closed and bounded

Bevis 45: 3.5.4

We have that

- A: K compact"
- B: K closed"
- X: K bounded"

The proposition tells

$$A \Rightarrow (B \wedge C)$$

which translates to

$$A \Rightarrow B \ \wedge \ A \Rightarrow C$$

K is compact if $\forall (x_n) \in K \ \exists (x_{n_k}) \in K$ such that $d(x_{n_k}, x) \to 0$ for $x \in K$. To prove that a set is compact, we thus have to show that all sequences have subsequences converging to a point in the set. Its easier to show that there exists a single sequence that doesnt have a subsequence converging to the point x . So we prove

$$\neg B \Rightarrow \neg A \ \neg C \Rightarrow \neg A$$

Start with $\neg B \Rightarrow \neg A$. Assume that K is not closed. That means that $\exists (x_n) \in K$ such that $\lim_{n\to\infty} x_n = x \notin K$. In english, this means that there is a boundary point $x \notin K$. Take any $(x_n) \in K$ converging to x. Then $\lim_{n\to\infty} x_n = x$. And since x_n converges to a point x, then so does all its subsequences. So

$$\lim_{n\to\infty} x_{n_k} = x$$

But since K is not closed, there exists limit points that are not in K. So $x \notin K$, and thus, there exists sequences in K such that there are no subsequences converging to a point in K. Hence, K cannot be compact if K is not closed

$\neg C \Rightarrow \neg A$ Assume now that K is not bounded, that is

$$\forall M \in \mathbb{R} \ \exists a, b \in K \text{ such that } d(a, b) > M$$

Pick $x_0 \in K$. Since $\forall n \in \mathbb{N} \ \exists x_n \in K$ such that $d(x_n, x_0) > n$, then

$$\lim_{n\to\infty} d(x_n, x_0) = \infty$$

We want to show that x_n has no convergent subsequence. This happens if for $x \in K$, we have that

$$\lim_{k\to\infty} d(x_{n_k}, x) = 0$$

And from the triangle ineq.

$$d(x_{n_k}, x_0) \leqslant d(x_{n_k}, x) + d(x, x_0)$$

Since $d(x_{n_k}, x) \to 0$ as $k \to \infty$, we have that $d(x_{n_k}, x_0) \leqslant d(x, x_0) =: C$. But from above, we have that

$$d(x_{n_k}, x_0) > n_k$$

and as $k \to \infty$ we get that $d(x_{n_k}, x_0) \to \infty$. And hence its a contradiction, since

$$\infty \leqslant C$$

is not true.

Korollar 1: 3.5.5

A subset of $\mathbb{R}^n$ is compact $\iff$ it is closed and bounded

Bevis 46: 3.5.5

We have already proved that $A \Rightarrow B\&C$, must now prove that a closed and bounded subset of $\mathbb{R}^n$ is compact.

Assume A is closed and bounded. From Bolsano Weierstrass theorem, any bounded sequence in $\mathbb{R}^n$ has a convergent subsequence. This means if we take $(x_n) \in A$, then

$$\lim_{k \to \infty} x_{n_k} = x$$

This can be viewed as yet another sequence. Define $(y_k) = (x_{n_k})$. Then since A is closed, $x \in K$, and we have found that if we pick an arbitrary sequence in $A \subseteq \mathbb{R}^n$, then we get that $\exists\, (x_{n_k})$ such that $d\,(y_k, x) \to 0$ as $k \to \infty$

Lemma 3: 3.5.6

Assume that $\{x_n\}$ is a caucht sequence in a metric space. If there is a subsequence $\{x_{n_k}\}$ converging to a point a, then the original sequence $\{x_n\}$ also converges to a.

Bevis 47: 3.5.6

Assume $(x_n) \in (X, d)$ is cauchy, and that (x_n) has a subsequence (x_{n_k}) such that $d\,(x_{n_k}, a) \to 0$ as $k \to \infty$. Since (x_n) is cauchy, we have that $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $d\,(x_n, x_m) < \varepsilon\ \forall n, m \geqslant N$. Since (x_{n_k}) converges to a, there is a $K \in \mathbb{N}$ such that $n_K \geqslant N$ and $d\,(x_{n_k}, a) < \varepsilon$. Then

$$d\,(x_n, a) \leqslant d\,(x_n, x_{n_K}) + d\,(x_{n_K}, a) < \varepsilon + \varepsilon = 2\varepsilon$$

Proposisjon 18: 3.5.7

Every comapct metric space is complete

Bevis 48: 3.5.7

Assume that K is compact. Then $\forall\,(x_n)\ \exists\,(x_{n_k})$ such that (x_{n_k}) converges to $x \in K$. K is complete is all cauchy sequences converge.

Assume now that (x_n) is a cauchy sequence in compact metric space (X, d). Since X is compact, (x_n) has convergent subsequences. And from the lemma above, if a cauchy sequence has convergent subsequences, then (x_n) also converges to $x \in X$. Hence all cauchy sequences converge, and X is complete as well

Proposisjon 19: 3.5.8

A closed subset F of a compact set K is compact

Bevis 49: 3.5.8

Assume F is closed. Then all convergent subsequences in F converges to $a \in F$. Assume (x_n) is a sequence in F. Since K is compact, every sequence (x_n) has convergent subsequences (x_{n_k}) to $a \in L$.

But since F is closed, any sequence must stay within the closed set. which means that any subsequence must also stay within F. Then $a \in F$, and thus, (x_{n_k}) converges to $a \in F$, which means that an arbitrary sequence $(x_n) \in F$ has a convergent subsequence (x_{n_k}) converging to $a \in F$, which means that F is also compact.

Proposisjon 20: 3.5.9

Assume that $f : X \to Y$ is a continuous function between two metric spaces. If $K \subseteq X$ is compact, then $f\,(K)$ is a compact subset of Y

Bevis 50: 3.5.9

Let (y_n) be a sequence in $f(K)$. We have that $K \subseteq X$ is compact. Since $f(x) \in f(K)$, we have for each $n \in \mathbb{N}$ $f(x_n) = y_n$. Since K is compact, (x_n) has a converging subsequence (x_{n_k}) to $x \in K$. But since f is a continuous function, then taking the function of a convergent sequence will preserve convergence in the new metric space. Thus, $f(x_{n_k})$ converges to $f(x)$

$$(y_{n_k}) = (f(x_{n_k}))$$

and thus, there is a subsequence of (y_n) converging to $y = f(x)$, and Y is compact as a result

Teorem 2: 3.5.10 Extreme value theorem

Assume K is nonempty compact subset of (X, d) and that $f : K \to \mathbb{R}$ is a continuous function. Then f has a maximum and minimum points in K, i.e. $\exists c, d \in K$ such that

$$f(d) \leqslant f(x) \leqslant f(c) \ \forall x \in K$$

Bevis 51: 3.5.10

Since K is compact, its closed and bounded. Define now

$$\alpha := \inf\{x \in K : f(x)\} \ \beta := \sup\{x \in X : f(x)\}$$

then

$$\alpha \leqslant f(x) \leqslant \beta \ \forall x \in K$$

Definisjon 10: 3.5.11

A subset A of X is called totally bounded if for each $\varepsilon > 0$ there is finite $B(a_1, \varepsilon), V(a_2, \varepsilon), ..., B(a_n, \varepsilon)$ of balls with centers in A and radius ε that cover A

$$A \subseteq B(a_1, \varepsilon) \cup V(a_2, \varepsilon) \cup ... \cup B(a_n, \varepsilon)$$

Proposisjon 21: 3.5.12

Let K be a compact subset of X. Then K is totally bounded

Bevis 52: 3.5.12

Assume that K is not totally bounded. This means that $\exists \varepsilon > 0$ such that no finite collection of ε- balls cover K. We shall make a sequence (x_n) that has no convergent subsequence.

Pick $x_1 \in K$, then $B(x_1, \varepsilon)$ does not cover K. Choose $x_2 \in K \backslash B(x_1, \varepsilon)$, then $B(x_1, \varepsilon) \cup B(x_2, \varepsilon)$ does not cover K. So we pick $x_3 \in K \backslash (B(x_1, \varepsilon), B(x_2, \varepsilon))$, and get that $B(x_1, \varepsilon) \cup B(x_2, \varepsilon) \cup B(x_3, \varepsilon)$ also does not cover K.

Go down and get that $x_n \in K \backslash (B(x_1, \varepsilon) \cup ... \cup B(x_{n-1}, \varepsilon))$. Now we have a sequence (x_n) such that $d(x_n, x_m) \geqslant \varepsilon \ \forall n, m \in \mathbb{N}$ and $n \neq m$. Hence (x_n) has no convergent subsequence, and A is not compact.

Teorem 3: 3.5.13

A subset A of complete metric space X is compact $\iff$ A is closed and totally bounded

Bevis 53: 3.5.12

To be continued

Oppgave 25: 3.5.5

Let (X, d) be a metric space. For a subset $A \subset X$, let ∂A denote the boundary of A. $\bar{A} = A \cup \partial A$

- a) A subset A of X is called precompact $\bar{A}$ is compact. Show that A is precompact if and only if all sequences in A have convergent subsequences

Bevis 54: 3.5.5 a)

- 1: A is precompact
- 2: all sequences in A have convergent subsequences.

We start with $1 \Rightarrow 2$ Assume that A is precompact. That is $\bar{A}$ is compact. Want to show that all sequences in A have convergent subsequences.

Take an arbitrary sequence in A, (x_n). Then $(x_n) \subset \bar{A}$, which is compact. This means that (x_n) has a convergent subsequence (x_{n_k}) to $x \in \bar{A}$. Hence all sequences in A has a convergent subsequence

Now we show the other way $2 \Rightarrow 1$

Assume that all sequences in A have convergent subsequences. Let $(x_n) \subset A$. Then (x_{n_k}) converges to $x \in X$. But since $A \subseteq \bar{A}$, then $x \in \bar{A}$, and hence, since $\bar{A}$ is closed by definition (the set $\bar{A}$ contains its boundary points) (x_{n_k}) must converge to $x \in \bar{A}$. Thus, $\bar{A}$ is compact $\Rightarrow$ A is precompact

- b) Show that a subset of $\mathbb{R}^n$ is precompact if and only if its bounded

Bevis 55: 3.5.5 b)

We have that

- A: $A \subseteq \mathbb{R}^n$ is precompact
- B: $A \subseteq \mathbb{R}^n$ is bounded

$A \Rightarrow B$

Assume that $A \subseteq \mathbb{R}^n$ is precompact. Since $\bar{A} \subseteq \mathbb{R}^n$ is compact, then this implies that $\bar{A}$ is closed and bounded. We already know that $\bar{A}$ is closed, but now its bounded as well. So $\exists M \in \mathbb{R}$ such that $d(a,b) \leqslant M \ \forall a, b \in \bar{A}$. Furthermore, since $A \subseteq \bar{A}$, then M also encloses A, and we get that A is also bounded.

$B \Rightarrow A$

Assume that $A \subseteq \mathbb{R}^n$ is bounded. Then we know that $\bar{A} = A \cup \partial A$ is closed and bounded. In $\mathbb{R}^n$, this implies that $\bar{A}$ is also compact. So $\bar{A}$ is compact $\iff$ A is precompact

And the proof is finished

Oppgave 26: 3.5.6

Assume (X, d) is a metric space and $f : X \to [0, \infty)$ is a continuous function. Assume $\forall \varepsilon > 0 \ \exists K_\varepsilon \subseteq K$ such that $f(x) < \varepsilon$ when $x \notin K_\varepsilon$. Show that f has a maximum.

Bevis 56: 3.5.6

We have that $X = K_\varepsilon \cup (K_\varepsilon)^c$. We assume that $\forall \varepsilon > 0 \ \exists K_\varepsilon \subseteq X$ such that $f(x) < \varepsilon$ when $x \in (K_\varepsilon)^c$.

When $x \in K_\varepsilon$, then, since K_ε is compact and f is continuous, we get that there exists a $d \in K_\varepsilon$ such that $f(x) \leqslant f(d) \ \forall x \in K_\varepsilon$. But we are asked to find the maximum of f in the whole metric space x. So when $x \notin K_\varepsilon$, then $f(x) < \varepsilon$ and ε becomes the upper bound for f. Thus, we can define

$$\sup_{x \in X} f(x) = max\left(f(d), \varepsilon\right)$$

What is bigger between $f(d)$ and ε?

Pick ε_1 and define

$$\varepsilon_n = e^{-n}\varepsilon_1$$

Then

$$\lim_{n \to \infty} \varepsilon_n = 0$$

Thus, $\exists N \in \mathbb{N}$ such that $d(\varepsilon_n, 0) < df(d) \ \forall n \geqslant N$. So we have found that

$$\sup_{x \in X} = f(d) \ \forall n \geqslant N$$

Oppgave 27: 3.5.7

Let (X, d) be a compact metric space, and assume that $f : X \to \mathbb{R}$ is continuous when we give $\mathbb{R}$ the usual metric $d(x, y) = |x - y|$. Show that if $f(x) > 0 \ \forall x \in X$, then $\exists a \in \mathbb{R} \ f(x) > a \ \forall x \in X$.

Bevis 57: 3.5.7

Since X is compact and f is continuous, then there exists a $c \in X$ such that $f(c) \geqslant f(x) \ \forall x \in X$. Since $f(x) > 0 \ \forall x \in X$ then also $f(c) > 0$. Hence there exists $a \in \mathbb{R}$ such that

$$f(c) = a \leqslant f(x) \ \forall x \in X$$

Oppgave 28: 3.5.8

Assume that $f : X \to Y$ is continuous between metric spaces, and let K be a compact subset of Y. Show that $f^{-1}(K)$ is closed. Find an examole which shows that $f^{-1}(K)$ need not be compact

Bevis 58: 3.5.8

Since $K \subseteq Y$ is compact, K is closed and bounded. So all sequences $(y_n) \in K$ have convergent subsequences meaning that (y_{n_k}) converges to $y \in K$. Since f is continuous, we have that $\forall n \in \mathbb{N}$, we have $f^{-1}(y_n) = x_n$. This means that

$$f^{-1}(y_{n_k}) = x_{n_k}$$

Now taking the limits, we get that $\lim_{k\to\infty} f^{-1}(y_{n_k}) = f^{-1}(\lim_{k\to\infty} y_{n_k}) = f^{-1}(y) = x \in f^{-1}(K)$. Since this counts for all (y_n) in K, then $f^{-1}(K)$ is closed.

Oppgave 29: 3.5.9

Show that a totally bounded subset of a metric space is always bounded. Find and example of a bounded set in a metric space that is not totally bounded.

Bevis 59: 3.5.9

Assume $A \subseteq X$ is totally bounded, then if $\exists \varepsilon > 0$ such that for a finite collection of balls, $A \subseteq B(a_1, \varepsilon) \cup ... \cup B(a_n, \varepsilon)$.

Choose $\varepsilon > 0$. We have that $A \subseteq \cup_{i=1}^{n} B(a_i, \varepsilon)$. Pick an arbitrary $x_0 \in X$, then define $R := max_{i\in[1,n]} d(a_i, x_0) + 1$, thus $d(a_i, x_0) < R \ \forall i \in [1, n]$. Pick $x \in A$ such that $x \in B(a_i, \varepsilon)$, then $d(a_i, x) < \varepsilon$. we get

$$d(x, x_0) \leqslant d(x, a_i) + d(a_i, x_0) < \varepsilon + R$$

hence $x \in B(x_0, R)$

Oppgave 30: 3.5.10

Let $d(x, y) = |x - y|$ be the usual metric on $\mathbb{Q}$. Let $A = \{q \in \mathbb{Q} : 0 \leqslant q \leqslant 2\}$. Show that A is closed and totally bounded subset of $\mathbb{Q}$, but not compact.

Bevis 60: 3.5.10

A is not compact, since the sequence converging to $\sqrt{2}$ does not lie in $\mathbb{Q}$, and hence not in A. But its closed, because $A^c = \{q \in \mathbb{Q} : (-\infty, 0) \cup (2, \infty)\}$ is open. Its totally bounded since

Oppgave 31: 3.5.11

A metric space (X, d) is locally compact if for each $a \in X$ is an $r > 0$ such that the closed ball $\bar{B}(a, r) = \{x \in X : d(a, x) \leqslant r\}$ is compact.

- a) Show that $\mathbb{R}^n$ is locally compact.

Bevis 61: 3.5.11 a)

Take any $x \in \mathbb{R}^n$, then the closed ball around x with radius r is $\bar{B}(x,r)$. Show that $\bar{B}(x,r)$ is compact.

Since $\bar{B}(x,r) \subseteq \mathbb{R}^n$, we have to show that $\bar{B}(x,r)$ is compact. This happens if its closed and bounded. Its already closed by definition, so we only have to show its bounded.

BOUNDEDNESS: $\exists M \in \mathbb{R}$ such that $d(a,b) \leqslant M \ \forall a,b \in \bar{B}(x,r)$. Since $d(x,y) \leqslant r \ \forall y \in \bar{B}(x,r)$, we have a bounded set.

- b) Show that if $X = \mathbb{R}\backslash\{0\}$ and $d : X \to \mathbb{R}$ is defined as $d(x,y) := |x-y|$, then (X,d) is locally compact, but not complete.

Bevis 62: 3.5.11 b)

We know that its not complete since the sequence $x_n = \frac{1}{n}$ converges to $0 \notin \mathbb{R}\backslash\{0\}$. Its locally compact if $\forall x \in \mathbb{R}\backslash\{0\}$ $\exists \varepsilon > 0$ such that $\bar{B}(x,\varepsilon)$ is compact. This happens if its closed and bounded. Its already closed from definition, it contains its boudnary points.... Its bounded if $\exists M \in \mathbb{R}$ such that $d(x,y) \leqslant M \ \forall y,x \in \bar{B}(x,eps)$. So we have that $\forall y \in \bar{B}(x,\varepsilon) \Rightarrow d(x,y) = |x-y| \leqslant \varepsilon =: M$. Hence we have that its closes

Oppgave 32: 3.5.15

Assume C and K are disjoint compact subset of a metric space (X,d) and define

$$a = \inf\{d(x,y) : x \in C, y \in K\}$$

Show that $a > 0$ and that $\exists x_0 \in C, y_0 \in K$ such that $d(x_0,y_0) = a$

Bevis 63: 3.5.15

Since $C \subseteq X$ and $K \subseteq X$, then C and K inherits the metric, and (C,d_C) and (K,d_K) are two new metric spaces in (X,d) Construct a new metric space $c \times K$ with the metric $d_\times((x,y),(x',y')) = d_C(x,x') + d(y,y')$. Since the distance function $d_C : C \to \mathbb{R}$ and $d_K : K \to \mathbb{R}$ are continuous, then $f : C \times K \to \mathbb{R}$ defined by $f(x,y) = d_{C\times K}(x,y)$, then f is a continuous function. To show that $a > 0$, assume that $a = 0$. This means that $d(x,y) = 0 \Rightarrow x,y \in C \cap K = \varnothing$. So the statement that $d(x,y) = 0$ is false. Therefore, $a > 0$ (because $d(x,y) \geqslant 0$)

The product of two metric spaces are also compact, and hence, f attains a minimum

$$\exists x_0, y_0 \in C \times K \text{ such that } f(x_0,y_0) = min_{x,y \in C\times K} f(x,y)$$

We now just let $a = f(x_0,y_0)$ and get that $d(x_0,y_0) = a$

Bevis 64: Proof that the product of two compact metric spaces are also compact

Let (X, d_X) and (Y, d_Y) be two compact metric spaces. We have that

$$d_X : X \to \mathbb{R}$$

$$d_Y : Y \to \mathbb{R}$$

The product of the two metric spaces make up $(X \times Y, d_{X \times Y})$. Is this new metric space compact?

We have that since (X, d_X) and (Y, d_Y) are compact, then

$$(x_n, y_n)$$

have convergent subsequences

$$(x_{n_k}, y_n) \to (x, y_n)$$

and

$$(x_n, y_{n_k}) \to (x_n, y)$$

hence

$$d_{X \times Y}\left(\left(x_{n_k}, y_{n_k}\right), (x, y)\right) \to d_X\left(x_{n_k}, x\right) + d_Y\left(y_{n_k}, y\right) = 0$$

So $(x_{n_k}, y_{n_k}) \to (x, y)$ as $k \to \infty$

Oppgave 33: A metric space

Define a metric space as rigorously as possible

Bevis 65

A metric space (X, d_X) is a non-empty set that satisfies

- $d(x, y) \geqslant 0$ and $d(x, y) = 0 \iff x = y \ \forall x, y \in X$
- $d(x, y) = d(y, x) \ \forall x, y \in X$
- $d(x, y) \leqslant d(x, z) + d(z, y) \ \forall x, y, z \in X$

Oppgave 34: A subset of metric space is bounded

Bevis 66

$A \subseteq (X, d)$ is boudned if $\exists M \in \mathbb{R}$ s.t. $d(a, b) \leqslant M \ \forall a, b \in A$

Oppgave 35: An open subset of a metric space

Bevis 67

An open subset $A \subseteq X$ of a metric space (X, d) is open if and only if $\forall x \in A \ \exists \varepsilon > 0$ such that $B(x, \varepsilon) = \{y \in A : d(x, y) < \varepsilon\} \subseteq A$

Oppgave 36: A closed subset of a metric space

Write as rigorously as possible the definition

Bevis 68

A closed subset $A \subseteq X$ of metric space (X, d) is defined as

$\forall (x_n) \in A$ such that $(x_n) \to x \in X \Rightarrow x \in A$

Bevis 69: Alternative proof

A contains all the limit points of any convergent sequence. That is, if $x \in X$ satisfies

$$\forall \varepsilon > 0 \; B(x, \varepsilon) \cap (A \backslash \{x\}) = \varnothing$$

then $x \in A$

Bevis 70: Alternative

$A \subseteq X$ is closed $\iff$ $A^c = X \backslash A$ is open. That is, $\forall x \in X \backslash A \; \exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq X \backslash A$

Oppgave 37: Convergence of a sequence in a metric space

Bevis 71

$(x_n) \in X$ converges to x if $\forall \varepsilon > 0 \; \exists N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon \; \forall n \geqslant N$

Bevis 72

$(x_n) \to x$ as $n \to \infty$

Oppgave 38: A function f from one metric space to another is continuous at a point x

Bevis 73

Let (X, d_X) and (Y, d_Y) be two metric spaces. f is continuous at $x \in X$ if $\forall \varepsilon > 0 \; \exists \delta > 0$ such that $\forall y \in X \; d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon$

Bevis 74

$\forall (x_n) \to x \in X \; f(x_n) \to f(x)$

Oppgave 39: A sequence in a metric space is cauchy

Bevis 75

Let (X, d) be a metric space, $\forall \varepsilon > 0 \; \exists N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon \; \forall n, m \geqslant N$

Oppgave 40: A complete metric space

Bevis 76

A metric space (X, d) is complete if all cauchy sequences converge. That is, let (x_n) be a cauchy sequence in (X, d), then $\forall \varepsilon > 0 \; \exists N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon \; \forall n, m \geqslant N$